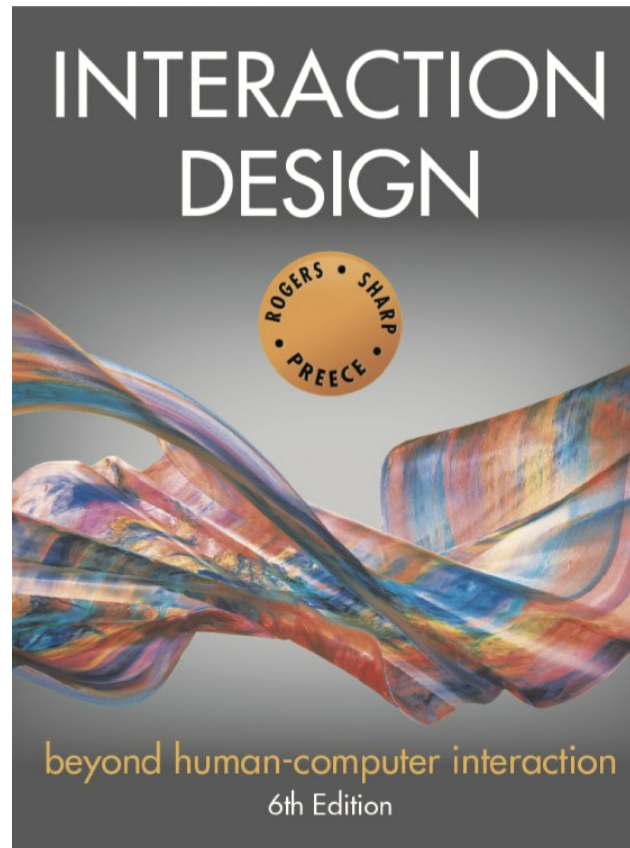




## Chapter 4

# COGNITIVE ASPECTS

# Overview

- What is cognition?
- Why it is important to understand in HCI
- Describe how cognition has been applied to interaction design
- Explain what are mental models and how to elicit them
- Cover relevant theories of cognition

# What is cognition?

- **Mental processes** involved in acquiring, processing, storing, and using **information**.
  - Thinking, remembering, learning, daydreaming, decision-making, seeing, reading, talking, writing...
- Ways of classifying cognition at a higher level:
  - Experiential vs. reflective cognition (Norman, 1993)
  - Fast vs slow thinking (Kahneman, 2011)

# Experiential vs. Reflective Cognition

- Experiential cognition:
  - Process by which people perceive, act, and react to events around them **intuitively** and **effortlessly**
  - Examples: driving a car, reading a book, having a conversation, and watching a video
- Reflective Cognition:
  - involves mental effort, attention, judgment, and decision-making
  - Examples: designing, learning, and writing a report
- Both modes are essential for everyday life

# Which involves fast vs slow thinking?

- $2 + 2 =$
- $21 \times 29 =$
- What color eyes do you have?
- How many colors are there in the rainbow?
- How many months in the year have 31 days?
- What is the name of the first school you attended?

# How can understanding cognition help HCI?

- Provides knowledge about what people can and cannot be expected to do
- Identifies and explains the nature and causes of problems that people encounter when using technology
- Provides theories, modeling tools, guidance, and methods that can lead to the design of better interactive products

# Cognitive processes

- Mental processes related to acquiring, processing, storing, and using information.
  - Attention
  - Perception
  - Memory
  - Learning
  - Reading, speaking and listening
  - Problem-solving, planning, reasoning and decision-making

# Attention

- Selecting things on which to concentrate at a point in time from the mass of stimuli around us
- Allows us to focus on information that is relevant to what we are doing
- Involves audio and/or visual senses
- Focused and divided attention
  - Enables us to be selective in terms of the mass of competing stimuli, but limits our ability to keep track of all events
- Design recommendation
  - Information at the interface should be structured to capture peoples' attention, for example, use perceptual boundaries (windows), color, reverse video, sound, and flashing lights

# Activity: Find the price for a double room at the Quality Inn in Pennsylvania

Pennsylvania  
Bedford Motel/Hotel: Crinaline Courts  
(814) 623-9511 S: \$118 D: \$120  
Bedford Motel/Hotel: Holiday Inn  
(814) 623-9006 S: \$129 D: \$136  
Bedford Motel/Hotel: Midway  
(814) 623-8107 S: \$121 D: \$126  
Bedford Motel/Hotel: Penn Manor  
(814) 623-8177 S: \$119 D: \$125  
Bedford Motel/Hotel: Quality Inn  
(814) 623-5189 S: \$123 D: \$128  
Bedford Motel/Hotel: Terrace  
(814) 623-5111 S: \$122 D: \$124  
Bradley Motel/Hotel: De Soto  
(814) 362-3567 S: \$120 D: \$124  
Bradley Motel/Hotel: Holiday House  
(814) 362-4511 S: \$122 D: \$125  
Bradley Motel/Hotel: Holiday Inn  
(814) 362-4501 S: \$132 D: \$140  
Breezewood Motel/Hotel: Best Western Plaza  
(814) 735-4352 S: \$120 D: \$127  
Breezewood Motel/Hotel: Motel 70  
(814) 735-4385 S: \$116 D: \$118

One way to structure the same information

## Activity: Find the price of a double room at the Holiday Inn in Columbia

| South Carolina |                 |           |          |        |        |
|----------------|-----------------|-----------|----------|--------|--------|
| City           | Motel/Hotel     | Area code | Phone    | Rates  |        |
|                |                 |           |          | Single | Double |
| Charleston     | Best Western    | 803       | 747-0961 | \$126  | \$130  |
| Charleston     | Days Inn        | 803       | 881-1000 | \$118  | \$124  |
| Charleston     | Holiday Inn N   | 803       | 744-1621 | \$136  | \$146  |
| Charleston     | Holiday Inn SW  | 803       | 556-7100 | \$133  | \$147  |
| Charleston     | Howard Johnsons | 803       | 524-4148 | \$131  | \$136  |
| Charleston     | Ramada Inn      | 803       | 774-8281 | \$133  | \$140  |
| Charleston     | Sheraton Inn    | 803       | 744-2401 | \$134  | \$142  |
|                |                 |           |          |        |        |
| Columbia       | Best Western    | 803       | 796-9400 | \$129  | \$134  |
| Columbia       | Carolina Inn    | 803       | 799-8200 | \$142  | \$148  |
| Columbia       | Days Inn        | 803       | 736-0000 | \$123  | \$127  |
| Columbia       | Holiday Inn NW  | 803       | 794-9440 | \$132  | \$139  |
| Columbia       | Howard Johnsons | 803       | 772-7200 | \$125  | \$127  |
| Columbia       | Quality Inn     | 803       | 772-0270 | \$134  | \$141  |
| Columbia       | Ramada Inn      | 803       | 796-2700 | \$136  | \$144  |
| Columbia       | Vagabond Inn    | 803       | 796-6240 | \$127  | \$130  |

Another way to structure the same information

# Findings based on activity

- Tullis (1987) found that the two screens produced quite different results
  - 1st screen: Took an average of 5.5 seconds to search
  - 2nd screen: Took 3.2 seconds to search
- Why, since both displays have the same density of information (31percent)?
- Spacing
  - In the 1st screen, the information is bunched up together, making it hard to search
  - In the 2nd screen, the characters are grouped into vertical categories of information making it easier

# Information Presentation

The way information is displayed can greatly influence

- how easy or difficult it is to comprehend appropriate piece of information.

# Multitasking and attention

- Is it possible to perform multiple tasks without one or more of them being detrimentally affected?
- Multitasking can cause people to lose their train of thought, make errors, and need to start over
- Ophir et al. (2009) compared heavy vs light multitaskers
  - Heavy multitaskers were more prone to being distracted than those who infrequently multitask
  - Heavy multitaskers are easily distracted and find it difficult to filter irrelevant information

# Multitasking experiment

- Lotteridge et al. (2015) conducted another study involving writing an essay under two conditions: relevant or irrelevant information
  - Heavy multitaskers were easily distracted but able to put this to good use if the distracting sources were relevant to the task in hand
  - Irrelevant information was found to impact task performance negatively
- In sum
  - The main reason why multitasking is thought to be detrimental for human performance is that it overloads people's capacity to focus

# Multitasking at work

It is increasingly common for workers to multitask

- For example, hospital workers have to attend to multiple screens in an operating room that provide new kinds of real-time information
- This requires clinician's constant attention to check if any data is unusual or anomalous
- Need to develop new attention and scanning strategies

# Apps to help people refocus

- Apps have been designed to help people get back on track or avoid being distracted
  - many are designed to block or limit the distracting sources, such as notifications, newsfeeds and social media
- An example is FocusMe that claims to “wall off online temptation”
  - help people improve their willpower so they can develop better digital habits

# Is it OK to use a phone when driving?



How distracting is it to be texting on the phone while driving?

# No!

- Driving is very demanding and drivers are prone to being distracted and causing accidents
  - drivers' reaction times are longer to external events when talking on the phone in a car (Caird et al., 2018)
- Drivers often try to imagine what the other person's face is like – the person to whom they are speaking
  - Doing so competes with the processing resources needed to enable them to notice and react to what is in front of them
  - when using their phones they rely more on their expectations about what is likely to happen next
- But some drivers persist despite the dangers!

# Are hands-free phones safer to use when driving?

- No, as same type of cognitive processing is happening when talking
- The same thing happens when talking with front seat passenger
  - But both can stop in mid-sentence if a hazard is spotted allowing the driver to switch immediately to the road
  - So, it's less dangerous talking to a front seat passenger than a remote person
  - A remote person on the end of a phone is not privy to what the driver is seeing and will carry on the conversation when there is a hazard
  - This makes it difficult for the driver to switch all their attention to the road

# Design implications for attention

- Context: Make information salient when it needs to be attended to at a given stage of a task
- Use techniques to achieve this:
  - For example, color, ordering, spacing, underlining, sequencing, and animation
- Avoid cluttering visual interfaces with too much information
- Consider designing different ways to support effective switching and returning to an interface

# Perception

- How information is acquired from the world and transformed into experiences
- Obvious implication is to design representations that are readily perceivable, for instance:
  - Text should be legible
  - Icons should be easy to distinguish and read

# Is color contrast good? Find Italian

Black Hills Forest  
Cheyenne River  
Social Science  
South San Jose  
Badlands Park  
Juvenile Justice

Peters Landing  
Public Health  
San Bernardino  
Moreno Valley  
Altamonte Springs  
Peach Tree City

Jefferson Farms  
Psychophysics  
Political Science  
Game Schedule  
South Addition  
Cherry Hills Village

Devlin Hall  
Positions  
Hubard Hall  
Fernadino Beach  
Council Bluffs  
Classical Lit

Results and Stats  
Thousand Oaks  
Promotions  
North Palermo  
Credit Union  
Wilner Hall

Highland Park  
Manchesney Park  
Vallecito Mts.  
Rock Falls  
Freeport  
Slaughter Beach

Creative Writing  
Lake Havasu City  
Engineering Bldg  
Sports Studies  
Lakewood Village  
Rock Island

Sociology  
Greek  
Wallace Hall  
Concert Tickets  
Public Radio FM  
Children's Museum

Performing Arts  
Italian  
Coaches  
McKees Rocks  
Glenwood Springs  
Urban Affairs

Rocky Mountains  
Latin  
Pleasant Hills  
Observatory  
Public Affairs  
Heskett Center

Deerfield Beach  
Arlington Hill  
Preview Game  
Richland Hills  
Experts Guide  
Neff Hall

Writing Center  
Theater Auditions  
Delaware City  
Scholarships  
Hendricksville  
Knights Landing

McLeansboro  
Experimental Links  
Graduation  
Emory Lindquist  
Clinton Hall  
San Luis Obispo

Brunswick  
East Millinocket  
Women's Studies  
Vacant  
News Theatre  
Candlewood Isle

Grand Wash Cliffs  
Indian Well Valley  
Online Courses  
Lindquist Hall  
Fisk Hall  
Los Padres Forest

Modern Literature  
Studio Arts  
Hughes Complex  
Cumberland Flats  
Central Village  
Hoffman Estates

# Are borders and white space better?

## Find French

Webmaster  
Russian  
Athletics  
Go Shockers  
Degree Options  
Newsletter

Curriculum  
Emergency (EMS)  
Statistics  
Award Documents  
Language Center  
Future Shockers

Student Life  
Accountancy  
McKnight Center  
Council of Women  
Commute  
Small Business

Dance  
Gerontology  
Marketing  
College Bylaws  
Why Wichita?  
Tickets

Geology  
Manufacturing  
Management  
UCATS  
Alumni News  
Saso

Intercollegiate  
Bowling  
Wichita Gateway  
Transfer Day  
Job Openings  
Live Radio

Thinker & Movers  
Alumni  
Foundations  
Corbin Center  
Jardine Hall  
Hugo Wall School

Career Services  
Doers & Shockers  
Core Values  
Grace Wilkie Hall  
Strategic Plan  
Medical Tech

Educational Map  
Physical Plant  
Graphic Design  
Non Credit Class  
Media Relations  
Advertising

Beta Alpha Psi  
Liberal Arts  
Counseling  
Biological Science  
Duerksen Fine Art  
EMT Program

Staff  
Aerospace  
Choral Dept.  
Alberg Hall  
French  
Spanish

Softball, Men's  
McKinley Hall  
Email  
Dental Hygiene  
Tenure  
Personnel Policies

English  
Graduate Complex  
Music Education  
Advising Center  
Medical School  
Levitt Arena

Religion  
Art Composition  
Physics  
Entrepreneurship  
Koch Arena  
Roster

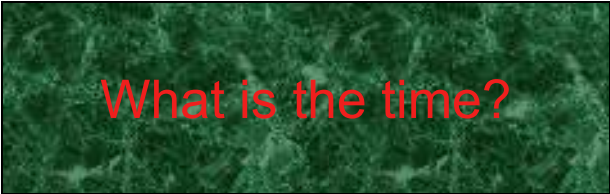
Parents  
Wrestling  
Philosophy  
Wichita Lyceum  
Fairmount Center  
Women's Museum

Instrumental  
Nursing  
Opera  
Sports History  
Athletic Dept.  
Health Plan

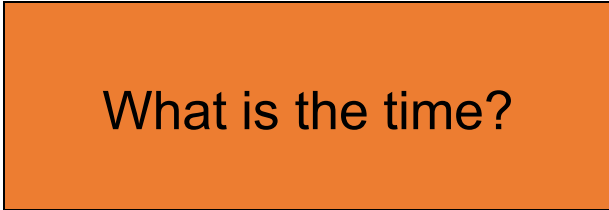
# Activity <sup>(1)</sup>

- Weller (2004) found people took less time to locate items for information that was grouped
  - Using a border (2nd screen) compared with using color contrast (1st screen)
- Some argue that too much white space on web pages is detrimental to search process
  - Makes it hard to find information
- Do you agree?

# Activity: Which is the easiest to read and why?



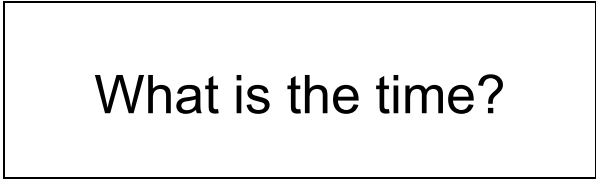
What is the time?



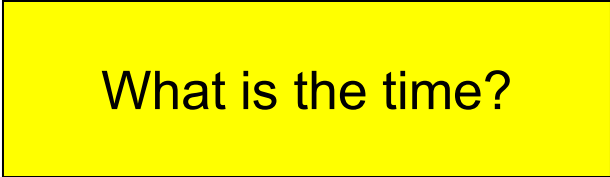
What is the time?



What is the time?



What is the time?



What is the time?

# Design implications <sup>(1)</sup>

- **Icons** should enable users to *distinguish* their meaning readily
- **Bordering** and **spacing** are effective visual ways of grouping information
- **Sounds** should be audible and distinguishable from each other
- Research proper **color contrast** techniques when designing an interface:
  - Yellow on black or blue is fine
  - Yellow on green or white is a no-no
- Haptic feedback should be used judiciously

# Memory

- Involves recalling various kinds of knowledge that allow people to act appropriately
  - For example, recognizing someone's face or remembering someone's name
- First **encode or store** and then **retrieve** knowledge
- We don't remember everything—it involves filtering and processing what is paid attention to
- Context is important as to how we remember (that is, where, when, how, and so on)
- We **recognize** things much better than being able to **recall** things
- We remember less about objects that we have photographed than when we observe them with the naked eye (Henkel, 2014)

# Recall

- The extent to which we can recall information depends on **two factors**
  1. The degree of **processing in the memory**
    - The more something is processed in the mind, the better we can recall
  2. **The context**
    - Context helps recall information better

# Processing in memory

- **Encoding** is first stage of memory
  - Determines which information is attended to in the environment and how it is interpreted
- The more **attention** paid to something...
- **The more it is processed** in terms of thinking about it and comparing it with other knowledge...
- **The more likely it is to be remembered**
  - For example, when learning about HCI, it is much better to reflect upon it, carry out exercises, have discussions with others about it, and write notes than just passively read a book, listen to a lecture or watch a video about it

# What is the difference between working memory and long-term memory?

- Working memory refers to our ability to recall a small amount of information from a recent time period
  - e.g. what someone said last during a conversation.
- Long-term memory is the capacity to recall memories from a longer time ago
  - e.g a tune someone heard from two decades ago

# Context is important

- Context affects the extent to which information can be subsequently retrieved
- Sometimes it can be difficult for people to recall information that was encoded or recorded in a different context:
  - “You are on a train and someone comes up to you and says hello. You don’t recognize him for a few moments, but then realize it is one of your neighbors. You are only used to seeing your neighbor in the hallway of your apartment building, and seeing him out of context makes him difficult to recognize initially”

# Activity <sup>(2)</sup>

- Try to remember the dates of your grandparents' birthday
- Try to remember
  - the cover of the last two books you read or
  - The image/graphic of the last app you downloaded
- Which was easiest? Why?
- People are very good at remembering visual cues about things
  - e.g. the color of items, the location of objects
- They find it more difficult to learn and remember arbitrary material, e.g. birthdays and phone numbers

# Recognition versus recall

- Command-based interfaces require people to recall from memory a name from a possible set of 100s of names
- Graphical interfaces provide visually-based options (menus, icons) that people need only browse through until they recognize one
- Web browsers provide tabs and history lists of visited URLs that support recognition memory

# Recognition versus Recall

- Command Line



```
>MONTH/08;DAY/21
```

- 2D menu to enable rapid selection



| August    |    |    |    |    |    |    |
|-----------|----|----|----|----|----|----|
| S         | M  | T  | W  | T  | F  | S  |
|           | 1  | 2  | 3  | 4  | 5  | 6  |
| 7         | 8  | 9  | 10 | 11 | 12 | 13 |
| 14        | 15 | 16 | 17 | 18 | 19 | 20 |
| <b>21</b> | 22 | 23 | 24 | 25 | 26 | 27 |
| 28        | 29 | 30 | 31 |    |    |    |

# The problem with the classic 7+or-2 memory phenomenon

- George Miller's (1956) theory of how much information people can remember
- People's immediate memory capacity is very limited to 7+or-2
- Has been applied in interaction design when considering how many options to display
- But is it a good use of a theory in HCI?
- Is it helpful?

# When creating an interface, should the designer...

- Present only 7 options on a menu
- Display only 7 icons on a tool bar
- Have no more than 7 bullets in a bullet list
- Place only 7 items on a pull down menu
- Place only 7 tabs on the top of a website page?
- Not necessarily...



# The reason is...

- People can scan lists of bullets, tabs, and menu items for the one they want
- They don't have to recall them from memory, having only briefly heard or seen them
- So you can have more than nine at the interface
  - For instance, history lists of websites visited
- Sometimes a small number of items is good
  - For example, smart watch displays
- Depends on task and available screen estate

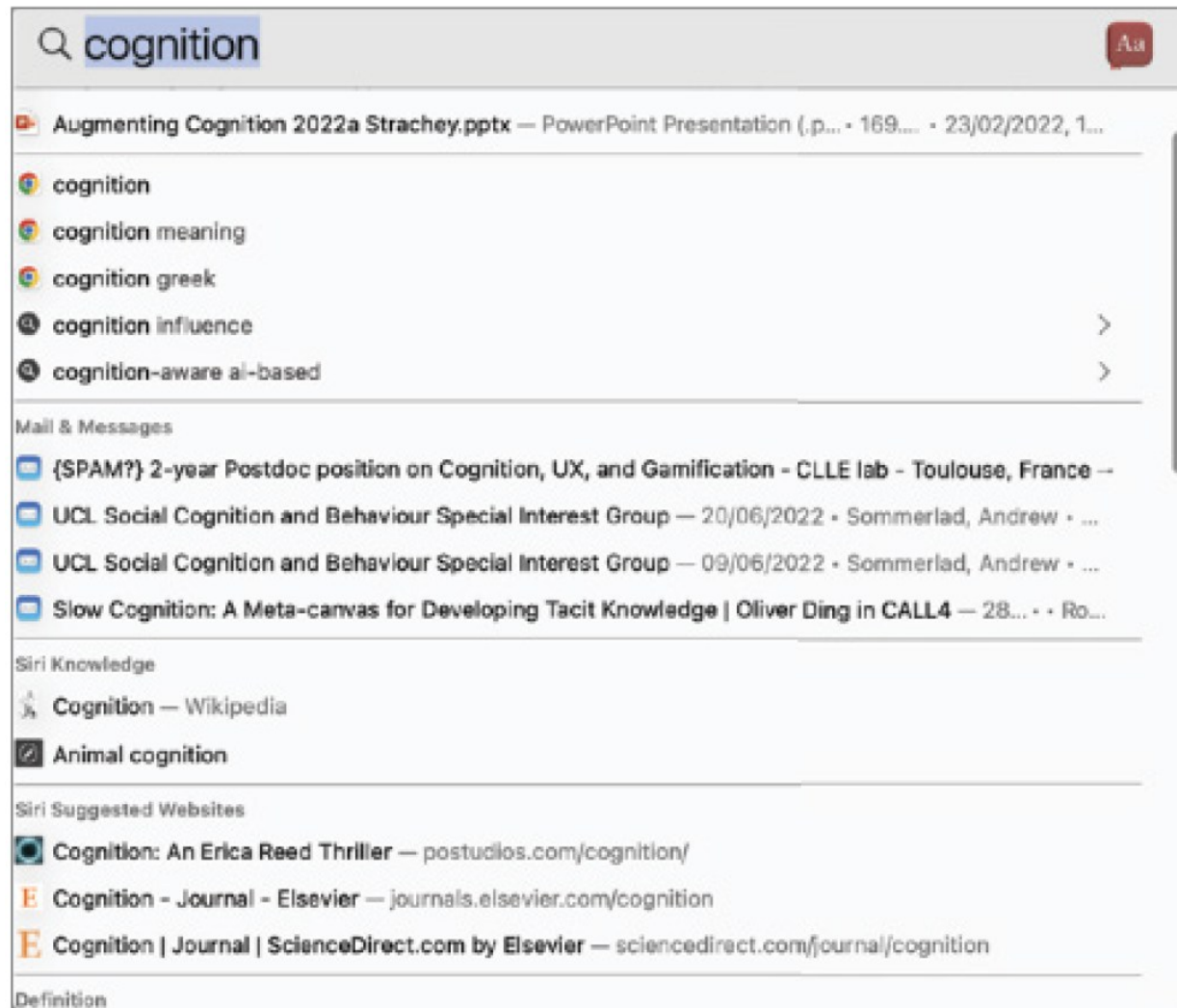
# Saving and searching files

- The number of documents written, images created, music files recorded, videoclips downloaded, emails with attachments saved, etc. is huge
  - Where and how to save them all?
  - Then remembering what they were called and where to find them again?
- Naming most common means of encoding them
- But can be difficult to remember, especially when you have 10,000s
- How might such a process be facilitated taking into account people's memory abilities?

# Personal Information management

- Bergman and Whittaker (2016) suggest helping people manage their “digital stuff” better
  - by deciding what personal information to keep
  - how to organize that information when storing it
  - which strategies to use to retrieve it later
- Provide richer metadata tools, such as time stamping, categorizing, tagging, and attribution (for example color, text, icon, sound, or image)
- However, trying to remember which metadata was created some time back may also prove to be difficult!
- What strategy do you find works for you best?

# Apple's Spotlight search tool



# Memory load

- Online, mobile and phone banking now require users to provide multiple pieces of information to access their account
  - For instance, ZIP code, birthplace, a memorable date, first school attended
  - Known as multifactor authentication (MFA)
- Why?
  - Increased security concerns
- Password managers, such as LastPass, have been developed that require only one main password
  - reduces stress and memory load on users
- Passwords could become extinct with the widespread use of biometrics and computer vision algorithms

# Digital Forgetting

- When might you wish to forget something that is online?
  - When you break up with a partner
  - Emotionally painful to be reminded of them through shared photos, social media, and so on.
- Sas and Whittaker (2013) suggest ways of harvesting and deleting digital content
  - For example, making photos of ex into an abstract collage
  - Helps with closure

# Memory aids

- SenseCam, developed by Microsoft Research Labs (now Autographer)
  - A wearable device that intermittently takes photos without any user intervention while worn
  - Digital photos taken are uploaded to a computer
    - Can be viewed later and help recall events associated with the photos
  - Has been found to improve people's memory, especially those suffering from dementia/Alzheimer's
- Other aids include RemArc, which triggers long-term memory using old BBC materials: photos, videos, sound clips

# SenseCam



The SenseCam device and a digital image taken with it

# Design implications <sup>(2)</sup>

- Reduce cognitive load by avoiding long and complicated procedures for carrying out tasks
- Design interfaces that promote **recognition rather than recall**
- Provide users with various ways of labelling digital information to help them easily identify it again
  - For example, folders, categories, color, tagging, and time stamping

# Learning

- Involves the accumulation of **skills** and **knowledge** and it **depends on memory**
- Two main types:
  - **Incidental learning** (for example, recognizing people's faces, what you did today)
  - **Intentional learning** (for instance, studying for an exam, learning to cook)
  - Intentional learning is much harder!
  - Many technologies have been developed to help (for example, multimedia, animations, VR)
- People find it hard to learn by following instructions in a manual
- People prefer to learn by doing

# Design implications <sup>(3)</sup>

- Design interfaces that support active learning through exploratory interaction
- Design interfaces that constrain and guide users
- Dynamically linking concepts and representations can facilitate the learning of complex material

# Reading, speaking, and listening

## Three forms/modes of language processing

The ease with which people can read, listen, or speak differs, depending on the **person**, **task** and the **context**:

- Many prefer listening to reading
- Differences between the 3 modes:
  - Reading can be quicker than speaking or listening
  - Listening requires less cognitive effort than reading or speaking
  - Dyslexics have difficulties understanding and recognizing written words

# Applications

- Voice user interfaces allow users to interact with them by asking questions
  - e.g. Google Voice, Siri, Alexa
- Speech-output systems use artificially-generated speech
  - e.g. written text-to-speech systems for the visually impaired
- Natural-language systems enable users to type in questions and give text-based responses
  - e.g. chatbots

# Design implications <sup>(4)</sup>

- Speech-based menus and instructions should be short
- Accentuate the intonation of artificially generated speech voices
  - They are harder to understand than human voices
- Provide opportunities for making text large on a screen

# Problem-solving, planning, reasoning, and decision-making

- All these processes involve *reflective* cognition
  - For example, thinking about what to do, what the options are, and the consequences
- Often involves conscious processes, discussion with others (or oneself), and the use of artifacts
  - Such as maps, books, pen and paper
- Reasoning:
  - It involves working through different scenarios and deciding which is best option/solution
- Weighing up alternatives

# Problem-solving, planning, reasoning, and decision-making

- Reasoning:
  - It involves working through different scenarios and deciding which is best option/solution
- Decision Making:
  - Weighing up alternatives

# Design implications <sup>(5)</sup>

- Provide information and help pages that are easy to access for people who wish to understand more about how to carry out an activity more effectively (for example, web searching)
- Use simple and memorable functions to support rapid decision-making and planning

# Dilemma

- The app mentality is making it worse for people to make their own decisions because they are becoming risk averse (Gardner and Davis, 2013)
  - Instead, they now rely on a multitude of apps
  - This makes them increasingly anxious
  - They are unable to make decisions by themselves
  - They need to resort to looking up info, getting other's opinions on social media, and comparing notes
- Do you agree?
- Did it happen to you when deciding which university/school to attend?

# Cognitive frameworks

- These are used to explain and predict user behavior at the interface
  - Based on theories of behavior
  - Focus is on mental processes that take place
  - Also use of artifacts and representations
- Most well known are:
  - Mental models
  - Gulfs of execution and evaluation
  - Distributed cognition
  - External and embodied cognition

# Mental models

- People develop an understanding of a system through learning about and using it
- This knowledge is sometimes described as a mental model:
  - **Shallow Mental Model:**
    - How to use the system (what to do next)
  - **Deep Mental Model:**
    - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

# More on mental models

- Craik (1943) described mental models as:
  - Internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes
  - Imagery and analogies are activated
- Deep versus shallow models
  - For example, how to drive a car and how it works

# Everyday reasoning and mental models

- (a) You arrive home on a cold winter's night to a cold house. How do you get the house to warm up as quickly as possible? Set the thermostat to be at its highest or to the desired temperature?
- (b) You arrive home starving hungry. You look in the fridge and find all that is left is an uncooked pizza. You have an electric oven. Do you warm it up to 375 degrees first and then put it in (as specified by the instructions) or turn the oven up higher to try to warm it up quicker?

# Heating up a room or oven that is thermostat-controlled

- Many people when asked (a) choose the first option
- Why?
  - They think it will heat the room up quicker
  - General valve theory, where 'more is more' principle is generalized to different settings (for instance, gas pedal, gas cooker, tap, radio volume)
  - But it is a wrong mental model for thermostats based on on-off switch model
- Many people when asked (b) choose the first option
  - Electric ovens work on the same principle as thermostats
- Most of us have erroneous mental models (Kempton, 1996)

# Erroneous mental models

- Lots of people hit the button for elevators and pedestrian crossings at least twice
  - Why? Think it will make the lights change faster or ensure that the elevator arrives!
- What kinds of mental models do users have for understanding how interactive devices work?
  - Poor, often incomplete, easily confusable, based on inappropriate analogies and superstition (Norman, 1983)

# How can UX be designed to help people build better mental models?

- Clear and easy to use instructions
- Appropriate tutorials and contextual sensitive guidance
- Provide online videos and chatbot windows when needing help
- Transparency: to make interfaces intuitive to use
- Affordances of what actions an interface allows
  - For example, swiping, clicking, or selecting

# Summary

- Cognition involves many processes including attention, memory, perception, and learning
- The way an interface is designed can greatly affect how well people can perceive, attend, learn, and remember how to do their tasks
- Theoretical frameworks, such as mental models and external cognition, provide ways of understanding how and why people interact with products
- This can lead to thinking about how to design better products

# Exercise

Write down how you think a contactless card or smartphone app like Ali Pay works

- What information is sent between the card/smartphone and the card reader when it is placed in front of it?
- What is the maximum amount you can pay for something using a contactless card, Ali Pay or Apple Pay?
- Why is there an upper limit?
- How many times can you use a contactless card or Apple/Google Pay in a day?
- What happens if you have two contactless cards in the same wallet/purse?
- What happens when your contactless card is stolen and you report it to the bank? What does the bank do?